

Incident Lifecycle Automation in ServiceNow

Project Documentation

This document consolidates the complete project into the five-phase structure shown in the provided reference: Ideation, Requirement Analysis, Project Design, Project Planning, and Project Development.

Executive Summary

Incident Lifecycle Automation in ServiceNow demonstrates an end-to-end IT Service Management process for a Corporate VPN connectivity incident. The solution connects Incident Management with Knowledge Management, Assignment & Escalation, Change Management, Child Incidents, Resolution, Knowledge Creation, and SLA Tracking.

Project Vision

The project aims to provide Service Desk and Level 2 support teams with a standardized process to create, classify, troubleshoot, escalate, investigate, change, resolve, document, and validate incidents.

Complete 5-Phase Project Structure

Phase	Major Contents
Phase 1 – Ideation Phase	Problem Identification; Proposed Solution; Project Vision; Expected Business Outcome
Phase 2 – Requirement Analysis	Business Objectives; Functional Requirements; Stakeholder Mapping; Scope Definition; Testing Requirements
Phase 3 – Project Design Phase	Application Design; Service and CI Design; Incident Lifecycle; Knowledge, Change, Child Incident and SLA Design; User/Role Design
Phase 4 – Project Planning Phase	Execution Roadmap; Milestones; Business Scenario; Test Scenario; Testing Strategy; Deployment Validation Plan
Phase 5 – Project Development Phase	Service Configuration; Service Offering; CI; Incident; Classification; Knowledge; Reassignment; Level 2; Change; Child Incident; Resolution; Knowledge Creation; Final Validation

1. Ideation Phase

The Ideation Phase establishes the project problem, proposed solution, project vision, and expected business outcome for a structured Incident Management lifecycle in ServiceNow.

1.1 Problem Identification

Organizations often receive IT incidents through emails, phone calls, or informal communication. Without a structured Incident Management process, incidents may be:

- Improperly classified
- Assigned to the wrong support team
- Delayed during investigation
- Difficult to track
- Escalated without proper documentation
- Resolved without recording the root cause
- Difficult to use for future knowledge creation

This creates SLA risks and reduces visibility into the overall incident lifecycle.

1.2 Proposed Solution

The proposed solution is to implement a structured and automated Incident Management lifecycle in ServiceNow.

The solution connects the following processes:

Incident Management → Knowledge Management → Assignment & Escalation → Change Management → Child Incidents → Resolution → Knowledge Creation → SLA Tracking

1.3 Project Vision

The project aims to provide Service Desk and Level 2 support teams with a standardized process to:

- Create incidents
- Classify and prioritize incidents
- Use knowledge articles for troubleshooting
- Escalate incidents to appropriate teams
- Create emergency changes when required
- Track child incidents
- Document cause and resolution
- Generate knowledge from resolved incidents
- Monitor SLAs and related records

1.4 Project Outcome

The expected outcome is an end-to-end Incident Management process that improves:

- Incident resolution time
- SLA compliance
- Incident classification
- Team collaboration
- Change traceability
- Knowledge reuse
- Service visibility

1.5 Scope at Ideation Level

- Service, service offering, and configuration item setup
- Incident creation and classification
- Knowledge investigation and attachment
- Reassignment and Level 2 escalation
- Emergency Change integration
- Child Incident management
- Incident resolution and knowledge creation
- SLA and related-record validation

2. Requirement Analysis Phase

This phase translates the project idea into business objectives, functional requirements, stakeholder requirements, scope, and testing expectations.

2.1 Business Objectives

1. Enable Service Desk agents to create and classify incidents.
2. Provide knowledge-based troubleshooting support.
3. Enable proper reassignment and escalation.
4. Allow Level 2 teams to investigate and resolve incidents.
5. Create emergency Change Requests directly from incidents.
6. Support child incident management.
7. Capture incident cause and resolution details.
8. Generate Knowledge Articles from resolved incidents.
9. Track SLAs and related records.

2.2 Functional Requirements

Incident Management

- Incident creation
- Caller selection
- Short description and description

- Category and subcategory
- Urgency and priority
- Service and service offering
- Configuration Item
- Assignment group
- Assigned to
- Work notes
- Watch list

Knowledge Integration

- Agent Assist
- Knowledge article search
- Article attachment to incident
- Helpful feedback
- Knowledge article creation from resolved incident

Assignment & Escalation

- Service Desk assignment
- Level 2 Network team assignment
- Agent reassignment
- Automatic clearing of Assigned to when assignment group changes

Change Integration

- Create Change Request from incident
- Emergency change selection
- Automatic field population
- Change-to-incident relationship
- Put incident On Hold while awaiting change

Child Incident Management

- Create child incidents
- Link child incidents to parent
- Track child incident status

Resolution

- Capture probable cause
- Capture resolution code
- Capture resolution notes
- Resolve parent incident
- Resolve child incident

SLA Management

- Task SLA visibility
- SLA tracking
- Related records validation

2.3 Stakeholder Requirements

Stakeholder	Role	Requirement
End Users	Report IT issues	Quick resolution and visibility
Service Desk Agents	L1 Support	Easy incident creation and classification
Level 2 Support	Technical Resolution	Complete incident information and ownership
Change Management Team	Change Control	Structured emergency change process
ServiceNow Administrator	Platform Administration	Maintain configuration and automation

2.4 Scope Definition

The requirements cover the records, workflows, users, integrations, and validation activities needed to demonstrate the incident scenario from creation to resolution.

2.5 Testing Requirements

- Incident creation
- Field classification
- Assignment
- Knowledge search
- Knowledge article attachment
- Reassignment
- Level 2 investigation
- SLA visibility
- Change creation
- Change linkage
- Child incident creation
- Resolution
- Knowledge creation
- Related records

2.6 Acceptance Criteria

The project should successfully demonstrate the required incident fields, team routing, knowledge usage, change relationship, child incident relationship, resolution documentation, knowledge creation, and SLA visibility.

3. Project Design Phase

The Project Design Phase defines the application modules, supporting records, incident lifecycle, integrations, and user responsibilities.

3.1 Application Design

- Incident Management
- Service Configuration
- CMDB
- Knowledge Management
- Change Management
- SLA Management
- Service Operations Workspace

3.2 Data and Configuration Design

These records are used during incident classification.

Record	Configuration
Service	Name: Remote Access; Operational Status: Operational
Service Offering	Name: Corporate VPN; Parent: Remote Access; Operational Status: Operational
Configuration Item	Name: VPN Gateway

3.3 Incident Lifecycle Design

Incident Creation ↓

Classification ↓

Knowledge Investigation ↓

Service Desk Assignment ↓

Level 2 Escalation ↓

Investigation ↓

Emergency Change if Required ↓

On Hold – Awaiting Change ↓

Resolution ↓

Knowledge Creation ↓

SLA & Related Record Validation

3.4 Integration Design

Incident → Knowledge

Agents use Agent Assist to search for relevant knowledge articles.

Incident → Change

When infrastructure changes are required, the agent creates an Emergency Change Request directly from the incident.

Incident → Child Incident

Recurring or related user issues can be tracked through child incidents.

Incident → Knowledge Creation

After resolution, the incident can be used as the source for a new Knowledge Article.

Incident → SLA

Task SLAs are associated with the incident to monitor resolution performance.

3.5 User Roles

User	Responsibility
Beth Anglin	Service Desk Agent
David Loo	Level 2 Network Support
Samantha Green	Incident Watch List
ServiceNow Administrator	Configuration and maintenance
Change Management Team	Emergency Change Management

3.6 Design Validation Points

- Supporting records are available before incident creation.
- Incident information can flow into knowledge, change, child incident, and SLA processes.
- Assignment changes support Level 2 investigation.
- Related records provide traceability.
- The resolved incident can become a reusable knowledge source.

4. Project Planning Phase

The Planning Phase defines the execution roadmap, milestones, business scenario, test scenario, testing strategy, and deployment validation plan.

4.1 Execution Roadmap

Stage	Milestone	Major Activity
Planning	Milestone 1	Service, Service Offering and CI setup
Development	Milestone 2	Incident Record Creation
Development	Milestone 3	Incident Classification
Development	Milestone 4	Knowledge Integration
Development	Milestone 5	Reassignment & Escalation
Development	Milestone 6	Level 2 Investigation
Development	Milestone 7	Emergency Change Creation
Development	Milestone 8	Child Incident Creation
Development	Milestone 9	Incident Resolution
Development	Milestone 10	Knowledge Creation
Validation	Final	SLA and Related Record Validation

4.2 Business/Test Scenario

A user is unable to connect to the Corporate VPN from home.

Field	Scenario Value
Short Description	Unable to connect to Corporate VPN from home office
Caller	Michael Hoefer
Channel	Phone
Category	Network
Subcategory	VPN
Urgency	2 - Medium
Service	Remote Access
Service Offering	Corporate VPN
Configuration Item	VPN Gateway
Initial Assignment	Service Desk
Escalation	Network team
Level 2 Finding	VPN-SRV-02 requires a restart
Change	Emergency Change Request to restart VPN-SRV-02
Resolution	VPN service restarted and incident resolved
Knowledge	Resolution converted into a Knowledge Article

4.3 Testing Strategy

- Incident creation
- Field classification
- Assignment
- Knowledge search
- Knowledge article attachment
- Reassignment
- Level 2 investigation
- SLA visibility
- Change creation
- Change linkage
- Child incident creation
- Resolution
- Knowledge creation
- Related records

4.4 Deployment Validation Plan

- Confirm parent incident is resolved.
- Confirm child incident is resolved.
- Confirm Change Request is linked.
- Confirm Knowledge Article is created.
- Confirm resolution and cause are documented.
- Confirm SLA is visible.
- Confirm Related Records are available.
- Confirm Activity Log contains the expected lifecycle history.

4.5 Deliverables

- Configured ServiceNow records and workflow
- Incident lifecycle implementation
- Emergency Change Request relationship
- Parent-child incident relationship
- Knowledge Article
- Validation evidence
- Final project documentation

5. Project Development Phase

This phase contains the actual ServiceNow implementation. The following activities implement the planned incident lifecycle.

Phase 1: Create New Service, Service Offering

Milestone 1: Create New Service

Go to All

Search → cmdb_ci_service.LIST

Click New

Fill details

: Name: Remote Access

: Operational status: Operational

Submit

Click Submit

Milestone 2: Create New Service Offering

Go to All

Search → service_offering.LIST

Click New

Fill details

: Name: Corporate VPN

: Parent: Remote Access

: Operational status: Operational

Submit

Click Submit

Phase 2: Incident Record Creation

Creating Incident Record

Open ServiceNow.

Impersonate Beth Anglin.

Navigate to: Service Operations Workspace > List > Incidents > Open

Click on New

Fill the details

: Short description: Unable to connect to Corporate VPN from home office

: Caller: Michael Hoefer

Save

Click Save

Phase 3: Incident Classification

Updating Incident Details

Stay in the same incident.

Go to the Details tab.

Update

: Description: Michael advised he was able to connect yesterday evening, but this morning received an authentication error. The Internet works fine.

: Channel: Phone

: Category: Network

: Subcategory: VPN

: Urgency: 2 - Medium

: Service: Remote Access

: Service Offering: Corporate VPN

: Configuration Item: ThinkStation S20 (any one)

: Compose section → Work notes: Classified and assigned incident.

Assign and Save

Click Assign to me

Save

Phase 4: Knowledge Integration

Agent Assist Verification

Click Agent Assist on the right side panel.

Ensure Knowledge Articles is selected.

Verify knowledge articles appear based on short descriptions.

Open any one article.

Click the 3 dots and Mark Article as Helpful.

Click Attach Article.

Give comment: Follow the Steps and Resolve the Incident.

Select Attach Article.

Return to incident. Verify: Knowledge article linked in Activity log; Helpful indicator marked.

Phase 5: Reassignment & Escalation

Watch List Update

Add Samantha Bordwell to the Watch list so she can track the incident and receive updates.

Add Beth Anglin to the Work notes list so she can follow the work notes and stay updated on the incident resolution.

This allows the incident to be assigned to another team for further resolution and tracking, while ensuring the required members can monitor the incident and its updates.

Note: If any of the mentioned members cannot be found, select another available member instead.

Reassignment

Update Assignment group to: Network

Verify the Assigned to field clears automatically.

Click Save.

Phase 6: Incident Tracking by Level 2

Investigation by Network Team

Impersonate David Loo.

Select 3 dots on Banner → Select Workspaces → Navigate to: Service Operations
Workspace >> Incidents >> Open.

Open the VPN incident.

Click Assign to me.

Verify: Description visible; Knowledge articles visible; SLA visible in Task SLAs (Related Records section).

Phase 7: Emergency Change Request Creation

Create Change

Update Configuration Item from ThickStation to PowerEdge.

Save.

Return to incident. Update State: On Hold; On Hold Reason: Awaiting Change.

Save.

Phase 8: Child Incident Creation

Create Child Incident

From Related Records tab → Select Child Incidents >> New.

Short Description: Unable to connect to Corporate VPN

: Caller: select any user

: Description: User receiving VPN authentication failure error.

Save. Make note of the incident number.

Return to parent incident >> Related Records.

Verify child incident listed.

Phase 9: Incident Resolution

Add Cause

On the Overview tab → Add Cause

: Probable Cause: PowerEdge service was suspended and required restart.

Save.

Add Resolution

Add Resolution

: Resolution Code: Workaround provided

: Resolution Notes: Restarted VPN-SRV-02 service as per emergency change request.

Save resolution.

Click Resolve

Confirm popup → Resolve.

Phase 10: Knowledge Creation

Create Knowledge Article

Click Create Knowledge.

Select IT as Knowledge Base.

Select article Template as Standard >> Next.

Verify all the details >> Save.

Go back to the parent record.

Open Related Records → verify created knowledge.

Phase 11: Final Validation

Final Validation

Verify Parent Incident = Resolved.
Verify Child Incident = Resolved.
Verify Change Request = Linked to Incident.
Verify Knowledge Article = Created.
Verify Resolution = Documented.
Verify Cause = Documented.
Verify SLA = Visible.
Verify Related Records = Available.
Verify Activity Log = Complete.

Phase 12: Testing & Deployment Validation

Testing & Deployment

This scenario validates:

Incident lifecycle end-to-end
Escalation process
Change integration
Knowledge creation
Child incident management
SLA tracking
Multi-team collaboration

5.1 Final Validation Matrix

Validation	Expected Result
Parent Incident	Resolved
Child Incident	Resolved
Change Request	Linked to Incident
Knowledge Article	Created
Resolution	Documented
Cause	Documented
SLA	Visible
Related Records	Available
Activity Log	Complete

5.2 Development Outcome

The development phase demonstrates the complete ServiceNow implementation from supporting-record configuration through incident creation, classification, knowledge integration, escalation, Level 2 investigation, emergency change, child incident, resolution, knowledge creation, and final validation.

End-to-End Incident Scenario

A user is unable to connect to the Corporate VPN from home.

Area	Details
Incident	Unable to connect to Corporate VPN from home office
Caller	Michael Hoefer
Channel	Phone
Category / Subcategory	Network / VPN
Urgency	2 - Medium
Service	Remote Access
Service Offering	Corporate VPN
Configuration Item	VPN Gateway
Initial Assignment	Service Desk
Level 2 Team	Network
Technical Finding	VPN-SRV-02 requires a restart
Change	Emergency Change Request to restart VPN-SRV-02
Resolution	VPN service restarted and incident resolved
Knowledge	Resolution converted into a Knowledge Article

Integrated Workflow

Incident Creation → Classification → Knowledge Investigation → Service Desk Assignment → Level 2 Escalation → Investigation → Emergency Change if Required → On Hold – Awaiting Change → Resolution → Knowledge Creation → SLA & Related Record Validation

Final Validation Checklist

Validation	Expected Result
Parent Incident	Resolved
Child Incident	Resolved
Change Request	Linked to Incident
Knowledge Article	Created
Resolution	Documented
Cause	Documented
SLA	Visible
Related Records	Available
Activity Log	Complete

Project Outcome

- Improved incident resolution time
- Improved SLA compliance
- Better incident classification
- Improved team collaboration
- Change traceability
- Knowledge reuse
- Better service visibility

Conclusion

The project demonstrates a complete Incident Management lifecycle in ServiceNow. Service Desk agents can log and classify incidents, leverage knowledge for faster resolution, escalate to Level 2 teams, initiate emergency changes, manage child incidents, and document final resolution. The integration between Incident, Change, and Knowledge modules supports structured issue management, improved SLA adherence, and better IT service delivery.